ONBRD-04: Cloud & SaaS Integrations

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 4 of 10 | **Created:** January 2026 | **Last Updated:** 10/23/2026

Extending Visibility Beyond OneAgent

While OneAgent provides deep application and infrastructure monitoring, many organizations need visibility into cloud services and SaaS platforms that can't run an agent. This notebook covers how to integrate AWS, Azure, GCP, and third-party SaaS tools into Dynatrace.

Table of Contents

1. Integration Overview
2. AWS Integration
3. Azure Integration
4. GCP Integration
5. Extensions Framework
6. Dynatrace Hub
7. Common SaaS Integrations
8. Verifying Integrations
9. Next Steps

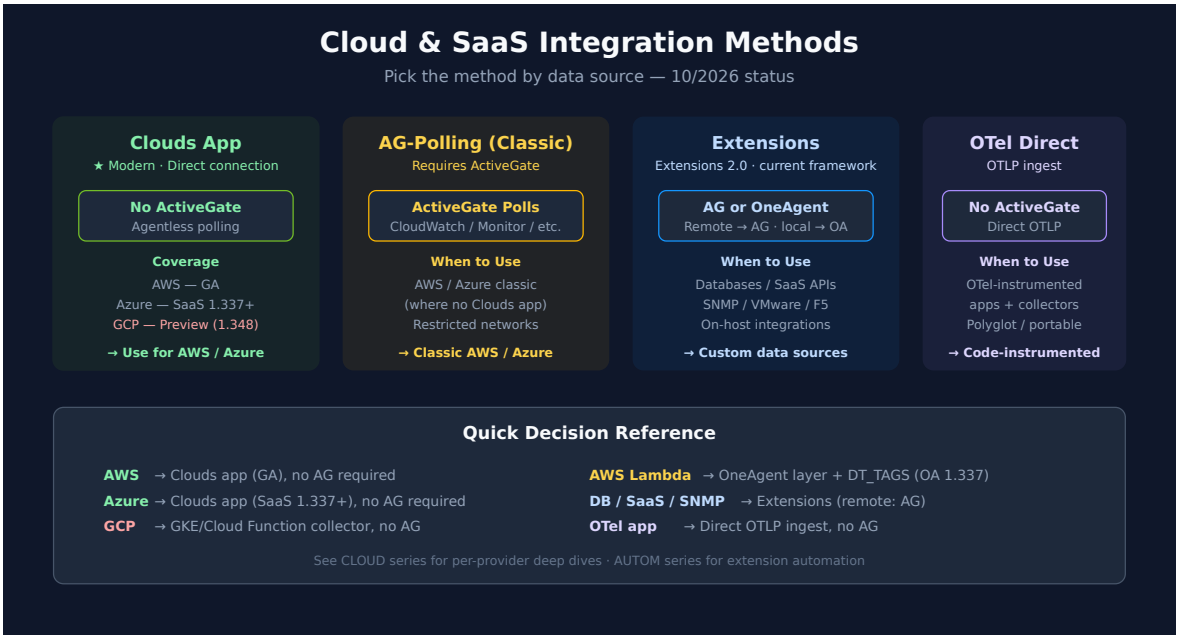
Prerequisites

- Dynatrace environment with admin access

- **ActiveGate** deployed where required — remote extensions and classic AWS / Azure polling run through an AG; AWS (Clouds app GA) and Azure (Clouds app, SaaS 1.337+) connections and the GCP integration on SaaS need none
- Cloud provider admin access (for AWS/Azure/GCP)
- API credentials for SaaS platforms

1. Integration Overview

Dynatrace offers multiple integration methods depending on the data source:



Method	Use Case	Requires ActiveGate?
Clouds App (recommended where supported)	AWS (GA), Azure (SaaS 1.337+); GCP listed as **Preview** and announced in SaaS 1.348 (rollout start 10/12/2026, staged tenant rollout) — verify it has reached your tenant	No (direct connection)
Cloud Integrations (Classic)	AWS / Azure where the Clouds app is not used (ActiveGate polling); GCP through the `dynatrace-gcp-monitor` collector on GKE or Cloud Functions	AWS / Azure: Yes — on SaaS hosted on AWS, Dynatrace provides one for the built-in AWS services · GCP: No on SaaS

Method	Use Case	Requires ActiveGate?
Extensions 2.0 *(current framework)*	Custom data sources, SaaS APIs, on-host integrations	Remote extensions: an ActiveGate group · local (on-host) extensions: OneAgent
OpenTelemetry	OTel-instrumented apps	No (direct OTLP ingest)
Log Ingest	External log sources	Optional (AG can route)
Metrics Ingest	Custom metrics via API	No

Decision Rule (Quick Reference)

Source	Default Method (10/2026)
AWS	Clouds app (GA) — direct connection
Azure	Clouds app — direct connection (SaaS 1.337+); *"no need to deploy ActiveGate compute resources for metric polling"* (Azure Cloud Platform Monitoring (DT docs))
GCP	Classic `dynatrace-gcp-monitor` (GKE or Cloud Function, no AG on SaaS) until the Clouds connection reaches your tenant; then evaluate it
AWS Lambda	Clouds app for CloudWatch metrics; OneAgent Lambda layer for traces and logs, with `DT_TAGS` primary fields (OneAgent 1.337+)
Custom DB / SaaS / SNMP	Extensions 2.0 — remote extensions on an AG group
OTel-instrumented app	OTLP ingest direct to Dynatrace

Why Integrate Before OneAgent?

Benefit	Description
Infrastructure context	See cloud resources before deploying agents
Dependency mapping	Understand managed services (RDS, Lambda, etc.)
Complete topology	Smartscape includes cloud services from day one

> **Sources:** SaaS 1.348 (DT docs) — *"There's no need to deploy ActiveGate compute resources for metric polling within your Google Cloud environment."*, OneAgent 1.337 (DT docs) — *"Primary fields set via DT_TAGS are now added to logs and spans in AWS Lambda deployments of OneAgent."*, Extensions (DT docs) — *"Run extensions remotely from an ActiveGate group to collect data from remote technologies and cloud environments."*
>
> **Where to go deeper:** the **CLOUD series** (9 notebooks) covers per-provider deep dives (AWS, Azure, GCP). The **AUTOM series** covers GitOps/Terraform automation for extension deployment at scale.

2. AWS Integration

The AWS integration pulls metrics from CloudWatch and discovers AWS resources.

Default path — Clouds app. Create the AWS connection in **Clouds**. *"All AWS connection creation methods are powered by CloudFormation as Infrastructure-as-Code (IaC) engine"*, so Dynatrace generates the stack that creates the access role, and the connection is *"fully managed by Dynatrace SaaS—no need to deploy ActiveGate compute resources for metric polling within your AWS environment."* See AWS Cloud Platform Monitoring (DT docs).

Supported Services

Category	Services
Compute	EC2, Lambda, ECS, EKS, Fargate
Database	RDS, DynamoDB, ElastiCache, DocumentDB
Storage	S3, EBS, EFS
Networking	ELB, ALB, NLB, API Gateway, CloudFront
Messaging	SQS, SNS, Kinesis, MSK
Other	Step Functions, Secrets Manager, and more

Classic Integration (Settings → Cloud and virtualization → AWS)

Use the classic integration only where the Clouds app is not used.

Authentication	When
IAM Role (role-based)	All new credentials outside AWS GovCloud and China
Access Key	AWS GovCloud and China partitions only — *"Key-based authentication is allowed only for AWS GovCloud and China partitions."*

On SaaS hosted on AWS, the classic integration needs no ActiveGate of your own for the built-in services; *"to monitor specific non-default AWS cloud services or if your AWS account exceeds 2,000 AWS resources, you must install and configure an Environment ActiveGate."*

IAM Role Setup (classic)

1. Create an IAM role with CloudWatch read permissions
2. Add trust relationship for Dynatrace's AWS account
3. Configure in Dynatrace: Settings → Cloud and virtualization → AWS

IAM policy: the first seven actions below are the ones the docs list as *"Permissions required for AWS monitoring integration"*. The rest are examples of service-specific permissions — take the complete per-service list from AWS CloudWatch metrics (DT docs) rather than this sample.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "cloudwatch:GetMetricData",
        "cloudwatch:GetMetricStatistics",
        "cloudwatch:ListMetrics",
        "sts:GetCallerIdentity",
        "tag:GetResources",
        "tag:GetTagKeys",
        "ec2:DescribeAvailabilityZones",
        "tag:GetTagValues",
        "ec2:DescribeInstances",
        "ec2:DescribeVolumes",
        "rds:DescribeDBInstances",
        "lambda:ListFunctions",
        "lambda:GetFunction"
      ],
      "Resource": "*"
    }
  ]
}
```

Configuration Location

Path: Settings → Cloud and virtualization → AWS

Setting	Recommendation
Services to monitor	Start with core services, expand as needed
Regions	Only regions where you have resources
Resource tags	Use tags to filter monitored resources

> **Sources:** AWS Cloud Platform Monitoring (DT docs), AWS CloudWatch metrics (DT docs) — *"Key-based authorization is no longer available for new credentials."*; the seven required actions are listed under *"Permissions required for AWS monitoring integration"*.

3. Azure Integration

The Azure integration uses Azure Monitor to collect metrics and discover resources.

Default path — Clouds app. *"Onboard your Azure subscriptions and turn them into native Dynatrace Azure connections, managing them from a dedicated Clouds app"* — add the Azure connection in **Clouds**, following Azure Cloud Platform Monitoring (DT docs). No ActiveGate is needed for metric polling.

Supported Services

Category	Services
Compute	Virtual Machines, App Service, Functions, AKS
Database	SQL Database, Cosmos DB, Redis Cache
Storage	Blob, Files, Queues, Tables
Networking	Load Balancer, Application Gateway, VNet
Messaging	Service Bus, Event Hubs, Event Grid

Classic Integration via App Registration

Use only where the Clouds app is not used:

1. Create an App Registration in Microsoft Entra ID (formerly Azure AD)
2. Grant **Reader** role on subscriptions to monitor
3. Create a client secret
4. Configure in Dynatrace with:
   - Tenant ID
   - Client ID
   - Client Secret
   - Subscription IDs

Configuration Location: Settings → Cloud and virtualization → Azure

4. GCP Integration

The GCP integration uses Cloud Monitoring (formerly Stackdriver) APIs.

> **Status note (10/2026):** Dynatrace's GCP setup page lists **GCP Cloud Platform Monitoring (Preview)**: *"Connect your Google Cloud accounts to Dynatrace and manage the newly created GCP connections entirely from Clouds."* SaaS 1.348 (rollout start 10/12/2026, staged tenant rollout) announces the new Clouds experience for Google Cloud: *"There's no need to deploy ActiveGate compute resources for metric polling within your Google Cloud environment."* Verify it has reached your tenant before planning around it. Until it does, the classic integration below is the working path. Review the **CLOUD series** for per-provider detail.

Supported Services

Category	Services
Compute	Compute Engine, GKE, Cloud Run, Cloud Functions
Database	Cloud SQL, Cloud Spanner, Firestore, Bigtable
Storage	Cloud Storage
Networking	Load Balancing, Cloud CDN
Messaging	Pub/Sub

Classic Integration: `dynatrace-gcp-monitor`

The classic integration is a collector you deploy inside Google Cloud. On SaaS it sends straight to your environment, so it needs **no ActiveGate**:

1. Deploy `dynatrace-gcp-monitor` with Helm on a GKE cluster — a new GKE Autopilot cluster is the recommended option — or deploy the metric integration as a Google Cloud Function.
2. Create the custom GCP role the setup page defines for the deployment (it covers GKE, Pub/Sub and related permissions, not just a read-only monitoring role).
3. Create a Dynatrace token from the **GCP Services Monitoring** template (the setup page creates it in **Access Tokens**).

4. Set `dynatraceUrl` to your environment: *"For SaaS log/metric ingestion, it's your environment URL (https://.live.dynatrace.com)."*

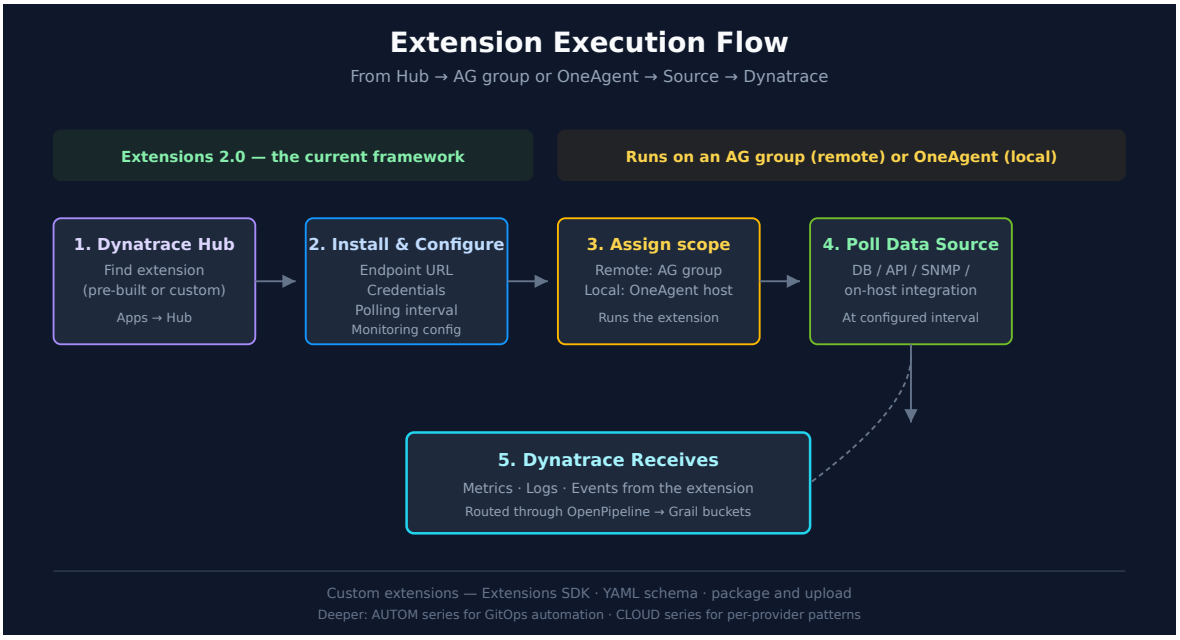
An ActiveGate appears in the classic setup only as a Managed option: *"For Managed deployments: You can use an existing ActiveGate for log ingestion."*

> **Sources:** Set up Dynatrace on Google Cloud (DT docs), SaaS 1.348 (DT docs), Google Cloud monitoring guide (DT docs), Set up the Dynatrace GCP integration on GKE (DT docs) — *"Under Template , select GCP Services Monitoring ."*

5. Extensions Framework

Extensions are Dynatrace's framework for integrating any data source — databases, SaaS platforms, network devices, on-host integrations, and more.

> **Recommendation for new customers:** build on **Extensions 2.0** — the current extensions framework. Extensions Framework 1.0 reached end of support on 2025-03-31 (Python EF1.0: 2024-10-31); JMX and PMI EF1.0 are deprecated (SaaS 1.347 adds an in-product banner) and reach end of support on SaaS on **07/01/2027** — *"As of July 1, 2027, all Extension Framework 1.0 JMX and PMI extensions will be out of support for SaaS Environments."* Migrate them to the Extensions 2.0 JMX / PMI extensions (EF1 JMX and PMI extensions end of support (DT docs)). Recent Dynatrace doc pages say simply "Extensions" and mean Extensions 2.0.



How Extensions Work

Component	Role
Extension Package	Code + metadata defining what to collect
ActiveGate group / OneAgent	Runs the extension — remotely from an ActiveGate group, or locally on the monitored host through OneAgent's Extension Execution Controller
Monitoring Configuration	Instance-specific settings (endpoints, credentials)
Dynatrace	Receives metrics, logs, events from the extension

Extension Categories

Category	Examples
Database	Oracle, SQL Server, PostgreSQL, MongoDB, Redis, Cassandra, MariaDB, DB2, HANA
On-host integrations	NGINX, HAProxy, Kafka, RabbitMQ, Elasticsearch, Memcached, Couchbase, Consul, Apache, etcd, Varnish, Zookeeper
Infrastructure	VMware, SNMP devices, F5, NetApp
SaaS	Salesforce, ServiceNow, Jira, Confluent

Category	Examples
Custom	Any REST API, custom protocols

Installing an Extension

1. **Find the extension** in Dynatrace Hub
2. **Install** to your environment
3. **Configure** with endpoint and credentials
4. **Assign** it — a remote extension to an ActiveGate group, a local extension to the hosts it should run on
5. **Verify** data is flowing

Monitoring Configuration

Each extension defines its own monitoring-configuration schema — the endpoint, credential and interval fields differ per extension — so there is no generic YAML to copy. Configure it from the extension's page in Hub, or through the Extensions 2.0 API: `POST /api/v2/extensions/{extensionName}/monitoringConfigurations` with a body of the documented shape, where `value` follows that extension's schema:

```
[
  {
    "scope": "HOST-D3A3C5A146830A79",
    "value": { }
  }
]
```

`scope` sets where the configuration runs (the example above is a host, as in the API reference); remote extensions run from the ActiveGate group you select. The API needs `extensions:configurations:write` (platform token) or `extensionConfigurations.write` (classic access token).

> **Sources:** Extensions (DT docs) — *"Run extensions locally on the monitored host to collect data from local data sources with the Extension Execution Controller."*, Extensions 2.0 API - POST a monitoring configuration (DT docs) — *"Platform Token / OAuth: Required scope: extensions:configurations:write"*

Creating Custom Extensions

For SaaS platforms without a pre-built extension, you can build a custom extension:

1. Use the Extensions 2.0 SDK
2. Define metrics schema in YAML
3. Write the polling logic
4. Package and upload to Dynatrace

SDK Documentation: Extensions Development

6. Dynatrace Hub

The Dynatrace Hub is your marketplace for extensions, integrations, and apps.

Accessing the Hub

Path: Apps → Dynatrace Hub

Or use quick search: **Cmd+K** → "Hub"

Hub Categories

Category	Examples
Cloud	AWS, Azure, GCP, Kubernetes
Database	Oracle, SQL Server, PostgreSQL
Infrastructure	VMware, SNMP, NetApp
Messaging	Kafka, RabbitMQ, IBM MQ
Observability	OpenTelemetry, Prometheus
Security	Snyk, SonarQube
ITSM	ServiceNow, Jira, PagerDuty

Installing from Hub

1. Search for the integration you need

2. Click **Install** or **Add to environment**
3. Follow the configuration wizard
4. Verify data appears in Dynatrace

7. Common SaaS Integrations

The methods below are typical starting points; Hub listings change, so confirm each platform's current listing in Hub before you plan.

Messaging Platforms

Platform	Integration Method	Key Metrics
Confluent Cloud	Extensions 2.0	Consumer lag, throughput, partitions
Amazon MSK	AWS Integration	Broker metrics, topic metrics
RabbitMQ	Extensions 2.0	Queue depth, message rates

CRM & Business Apps

Platform	Integration Method	Key Metrics
Salesforce	Extensions 2.0 / API	API calls, response times, limits
ServiceNow	Extensions 2.0	Incident metrics, workflow times

DevOps Tools

Platform	Integration Method	Key Metrics
GitHub	Webhooks	Deployment events, commit info
GitLab	Webhooks	Pipeline events, deployments
ArgoCD	Extensions 2.0	Sync status, app health

Database-as-a-Service

Platform	Integration Method	Key Metrics
MongoDB Atlas	Extensions 2.0	Connections, ops/sec, replication lag
AWS DocumentDB	AWS Integration	CloudWatch metrics
Snowflake	Snowflake integration in the Data Observability app (guided setup wizard); the Snowflake extension is deprecated	Query performance, warehouse usage

> **The Snowflake extension is deprecated.** Its documentation page says: *"The Snowflake extension is deprecated and has been replaced with the new Snowflake integration."* The replacement is the Data Observability app, which *"connects your Snowflake account with Dynatrace through a guided setup wizard."* Start new Snowflake monitoring there. Keep existing extension-based Snowflake configurations running until the integration is in place rather than tearing them down first. (A separate DSOA installation wizard in Business Observability arrived with SaaS 1.345.)
>
> **Sources:** Snowflake extension (DT docs), Data Observability app (DT docs), SaaS 1.345 (DT docs) — *"Introduced a Dynatrace Snowflake Observability Agent (DSOA) installation wizard that streamlines the setup process within Business Observability."*
>
> **Do not confuse this with the Snowflake *data connector*.** The connector installed from the Hub and configured under **Settings > Connections** powers Snowflake *workflow actions* (Execute Statement, Store Statement Result) — see the WFLOW series for that path. It is a different surface, serves a different purpose, and is unaffected by this change.

8. Verifying Integrations

After configuring integrations, verify data is flowing into Dynatrace.

```
// Check for AWS Lambda functions (modern Smartscape topology query)
smartscapeNodes "AWS_LAMBDA_FUNCTION"
| fields name, id
| limit 20

// Legacy alternative (deprecated dt.entity.* — still works on hybrid
tenants):
// fetch dt.entity.aws_lambda_function
// | fields entity.name, id
// | limit 20
```

```
// Check for Azure VMs (modern Smartscape topology query)
// Azure node types follow AZURE_&lt;RESOURCE_PROVIDER&gt;_&lt;TYPE&gt; -
"AZURE_VM" is not a node type and returns nothing.
smartscapeNodes "AZURE_MICROSOFT_COMPUTE_VIRTUALMACHINES"
| fields name, id
| limit 20

// Legacy alternative (deprecated dt.entity.* — still works on hybrid
tenants):
// fetch dt.entity.azure_vm
// | fields entity.name, id
// | limit 20
```

```
// Check for cloud-sourced metrics
//
// The `dt.cloud.aws.*` metric namespace DOES NOT EXIST in any spelling
(verified 08/12/2026).
// AWS CloudWatch metrics are published as
cloud.aws.&lt;service&gt;.&lt;CloudWatchName&gt;.By.&lt;Dimension&gt; —
// the CloudWatch name keeps its PascalCase and the split dimension is part of
the key. A
// timeseries against a missing key returns an EMPTY result rather than an
error, so this cell
// silently drew nothing. Enumerate with:
//   metrics | filter startsWith(metric.key, "cloud.aws") | fields metric.key
| sort metric.key asc
timeseries avgDuration = avg(cloud.aws.lambda.Duration.By.FunctionName),
from:-24h, by:{FunctionName}
| fieldsAdd avg_duration_ms = arrayAvg(avgDuration)
| fields FunctionName, avg_duration_ms
| sort avg_duration_ms desc
| limit 10

// Or browse available cloud metrics in the Dynatrace Metrics browser:
// Observe &amp;amp; Explore → Metrics → search for "cloud" or "aws" / "azure" /
"gcp"
```

```
// Check Extensions 2.0 status - one row per extension, status and ActiveGate
group.
// dt.extension.status: OK = collecting; UNASSIGNED = the configuration is not
bound to a
// running ActiveGate group or host; DATASOURCE_UNSUPPORTED = the data source
on that
// host/endpoint is not supported. No dt.active_gate.group.name = a local
extension
// running on OneAgent. datapoints = minutes reported in the window.
timeseries s = count(dt.sfm.extension.status), from:-24h,
    by:{dt.extension.name, dt.extension.status, dt.active_gate.group.name}
| fieldsAdd datapoints = arraySum(s)
| fields dt.extension.name, dt.extension.status, dt.active_gate.group.name,
datapoints
| sort dt.extension.status asc, dt.extension.name asc
| limit 50
```

Troubleshooting Integration Issues

Issue	Common Cause	Solution
No data appearing	Credentials invalid	Verify IAM role/service account
Partial data	Permissions incomplete	Check required permissions list
Delayed data	Polling interval	Cloud metrics arrive on the provider's publishing and polling cadence — compare against the provider console before treating it as a fault
Extension not running	Not assigned, or AG / host issue	Run the status query above — `UNASSIGNED` means no running AG group or host is bound; then check AG logs
Metrics but no entities	Tag configuration	Ensure resources are tagged correctly

9. Next Steps

With cloud and SaaS integrations configured:

1. **ONBRD-05: Deploying OneAgent** — Add deep application monitoring
2. **ONBRD-06: Organizing Your Environment** — Tag and segment integrated resources; design `dt.security_context`
3. Explore the topology in Smartscape to see cloud services
4. Create dashboards combining cloud metrics with application data

Where to Go Deeper

- **CLOUD series** (9 notebooks) — Per-provider integration deep dives (AWS, Azure, GCP)
- **AUTOM series** (14 notebooks) — GitOps / Terraform / Monaco automation for extension deployment
- **OPLOGS / OPMIG / OPIPE** — OpenPipeline routing for ingested cloud and SaaS data
- **OTEL series** — OpenTelemetry as the alternative ingest path

Integration Checklist

- [ ] AWS / Azure / GCP integration configured (Clouds app for AWS/Azure; `dynatrace-gcp-monitor` for GCP — no AG on SaaS — until the Clouds connection reaches your tenant)
- [ ] Required cloud permissions granted
- [ ] ActiveGate group assigned for remote extensions
- [ ] Extensions 2.0 evaluated for custom integrations; any remaining EF1.0 extensions (JMX / PMI: end of support on SaaS 07/01/2027) identified for migration
- [ ] Key SaaS platforms identified for integration
- [ ] Extensions installed from Hub
- [ ] Data verified flowing into Dynatrace

Summary

In this notebook, you learned:

- The integration method matrix (Clouds app / AG polling / Extensions / OTel / Ingest APIs)
- Per-provider status (AWS GA, Azure direct connection since SaaS 1.337, GCP Clouds connection in Preview; classic GCP needs no AG on SaaS)
- AWS, Azure, and GCP setup paths
- Extensions 2.0 as the current extensions framework
- Dynatrace Hub for discovery
- Common SaaS integration patterns
- DQL queries to verify integrations are landing in topology, and to check extension status

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References

- Cloud Platforms
- Clouds App

- AWS Monitoring
- Azure Monitoring
- Azure Cloud Platform Monitoring (DT docs) — the no-ActiveGate quote above
- What's new in Dynatrace SaaS 1.337 (DT docs)
- Set up Dynatrace on Google Cloud (DT docs)
- Set up the Dynatrace GCP integration on GKE (DT docs)
- AWS CloudWatch metrics (DT docs)
- Extensions Framework
- Dynatrace Hub

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.